

Curtin University, Semester 2, 2026

ECOM5005

Tutorial 7B Detailed Solutions

Confidence Interval Estimation - classroom-ready worked solutions

Learning Outcomes

1. Construct and interpret confidence intervals for means using Z or Student t as appropriate.
2. Construct and interpret confidence intervals for population proportions.
3. Explain the role of confidence level, standard error and margin of error.
4. Determine sample sizes for specified precision requirements.
5. Complete a beginner-friendly Excel confidence-interval analysis using a real-world business dataset.

PART 1 - Detailed worked solutions

Question 1. Solar-farm battery capacity

Question

Known $\sigma = 12$ kWh, $n = 64$, $\bar{x} = 84.6$ kWh. Construct 90%, 95% and 99% confidence intervals, compare their widths, and discuss the 80 kWh engineering target.

Detailed Solution

Step 1: Because the population standard deviation is treated as known, use the Z interval $\bar{x} \pm z^* \sigma / \sqrt{n}$. The standard error is $12 / \sqrt{64} = 12 / 8 = 1.5$ kWh.

(a) 90% confidence: $z^* = 1.645$. Margin of error = $1.645(1.5) = 2.4675$. CI = $84.6 \pm 2.4675 = (82.1325, 87.0675)$ kWh.

(b) 95% confidence: $z^* = 1.96$. Margin of error = $1.96(1.5) = 2.9400$. CI = $(81.6600, 87.5400)$ kWh.

(c) 99% confidence: $z^* = 2.576$. Margin of error = $2.576(1.5) = 3.8640$. CI = $(80.7360, 88.4640)$ kWh.

(d) Width = $2 \times$ margin of error. The widths are about 4.94, 5.88 and 7.73 kWh. Holding n and σ fixed, greater confidence requires a larger critical value, which makes the interval wider.

(e) Even the 99% interval has a lower endpoint around 80.74 kWh, above the 80 kWh target. The interval therefore supports the view that the population mean is above 80 kWh under the stated sampling/model assumptions. This is a confidence-interval statement, not a guarantee about every individual battery module.

Interpretation

The point estimate is 84.6 kWh. Precision falls as the confidence level rises: the 90% interval is narrowest and the 99% interval is widest. Higher confidence is purchased with a wider range of plausible values.

Answer Summary

90%: (82.13, 87.07). 95%: (81.66, 87.54). 99%: (80.74, 88.46). Standard error = 1.50 kWh.

Question 2. Airport baggage-drop processing time

Question

$n = 18$, $\bar{x} = 31.8$ seconds, $s = 6.4$ seconds, σ unknown. Construct and interpret a 95% confidence interval and evaluate the 35-second target.

Detailed Solution

Step 1: Because σ is unknown, use Student's t distribution. Degrees of freedom = $n - 1 = 17$. For 95% confidence, $t^* = t_{0.025, 17} \approx 2.1098$.

Step 2: Standard error = $s/\sqrt{n} = 6.4/\sqrt{18} = 1.5085$ seconds.

Step 3: Margin of error = $2.1098 \times 1.5085 = 3.1826$ seconds.

Step 4: CI = $31.8 \pm 3.1826 = (28.6174, 34.9826)$ seconds.

Step 5: The entire 95% interval lies below 35 seconds, although the upper endpoint is very close to 35. This supports the service target that the true mean processing time is below 35 seconds, provided the sampling assumptions are reasonable.

Step 6: Because $n = 18$ is small, the population distribution should be approximately normal and should not contain severe outliers. In practice inspect a boxplot and a normal probability (Q-Q) plot before relying heavily on the t interval.

Answer Summary

$df = 17$; $t^* \approx 2.1098$; $SE \approx 1.5085$; margin ≈ 3.1826 ; 95% CI = $(28.62, 34.98)$ seconds.

Question 3. Fintech customer retention

Question

Of $n = 250$ new subscribers, 146 remain active after 90 days. Estimate the population retention proportion with 95% confidence and assess the 60% marketing claim.

Detailed Solution

Step 1: Sample proportion $p = 146/250 = 0.584$.

Step 2: Check the large-sample conditions: $np = 250(0.584) = 146$ and $n(1-p) = 250(0.416) = 104$. Both are much greater than 5, so the normal approximation used in the lecture is appropriate.

Step 3: Standard error = $\sqrt{[p(1-p)/n]} = \sqrt{[0.584(0.416)/250]} = 0.03117$.

Step 4: For 95% confidence, $z^* = 1.96$. Margin of error = $1.96(0.03117) = 0.06110$.

Step 5: CI = $0.584 \pm 0.06110 = (0.52290, 0.64510)$, or 52.29% to 64.51%.

Step 6: The marketing claim of 60% lies inside the interval. Therefore 60% is compatible with the sample evidence; the interval does not provide a reason to rule it out.

Interpretation

We are 95% confident that the true 90-day retention proportion for the population represented by the random sample is between about 52.3% and 64.5%.

Answer Summary

$p = 0.584$; $SE = 0.03117$; 95% CI = (0.5229, 0.6451). The value 0.60 is inside the interval.

Question 4. Cold-chain audit sample-size planning**Question**

Use $\sigma = 18$ minutes. Find the sample sizes needed for margin ± 3 minutes at 95% and 99% confidence, then find the 95% margin if $n = 150$.

Detailed Solution

For a mean with planning standard deviation σ , $n = (z^*\sigma/e)^2$ and the result is always rounded UP.

(a) 95% confidence: $n = [1.96(18)/3]^2 = 138.30$. Round up: $n = 139$ shipments.

(b) 99% confidence: $n = [2.576(18)/3]^2 = 238.89$. Round up: $n = 239$ shipments.

(c) The 99% design needs more observations because its larger critical value must be offset by a smaller standard error if the same ± 3 -minute margin is to be maintained.

(d) With $n = 150$ at 95% confidence, $e = 1.96(18/\sqrt{150}) = 2.8805$ minutes. Thus 150 shipments would actually achieve slightly better than the ± 3 -minute precision target at 95% confidence.

Answer Summary

95%: $n = 139$. 99%: $n = 239$. With $n = 150$ at 95%, margin of error $\approx \pm 2.88$ minutes.

Question 5. AI-helpdesk escalation survey**Question**

Pilot $p = 0.18$; desired 95% margin ± 0.025 . Find required n using $p = 0.18$ and $p = 0.50$, and assess the margin if only 1,000 conversations are reviewed.

Detailed Solution

For a proportion, $n = z^{*2} p(1-p)/e^2$ and round UP. Here $z^* = 1.96$ and $e = 0.025$.

(a) Pilot $p = 0.18$: $n = (1.96^2 \times 0.18 \times 0.82)/(0.025^2) = 907.20$, so $n = 908$.

(b) Conservative $p = 0.50$: $n = (1.96^2 \times 0.50 \times 0.50)/(0.025^2) = 1536.64$, so $n = 1537$.

(c) The product $p(1-p)$ is maximized at $p = 0.50$. That produces the largest possible variance and therefore the largest required sample size for a fixed confidence level and margin of error.

(d) If $n = 1000$, using $p = 0.18$ gives $e = 1.96\sqrt{[0.18(0.82)/1000]} = 0.02381$, or about ± 2.38 percentage points. Using $p = 0.50$ gives $e = 1.96\sqrt{[0.25/1000]} = 0.03099$, or about ± 3.10 percentage points. Thus 1,000 is enough under the pilot-based design but not under the conservative worst-case design.

Answer Summary

Pilot $p = 0.18$: $n = 908$. Conservative $p = 0.50$: $n = 1537$. At $n = 1000$, margins are about $\pm 2.38\%$ and $\pm 3.10\%$, respectively.

PART 2 - Data exercise: Restaurant tips

Dataset context. The dataset contains 244 restaurant transactions recorded by one food server over about two and a half months. For teaching purposes, Sample_60 is a fixed random sample of 60 observations from that observed data frame. Use Sample_60 for the inferential calculations and only use Full_Data at the end as a benchmark.

Source: Bryant, P. G. & Smith, M. A. (1995), Practical Data Analysis: Case Studies in Business Statistics.

<https://iowabiostat.github.io/data-sets/tips/tips.html>

Beginner Excel workflow - follow these steps in order

Step 1: Open "Restaurant_Tips_Data.xlsx". Click Sample_60. The headers are in row 1 and the 60 observations are in rows 2:61. total_bill is column B, tip is column C, and smoker is column E. Then click Analysis_Workspace; enter your formulas in column B and interval endpoints in columns D and E.

Step 2: In Analysis_Workspace!B14 calculate n with =COUNT(Sample_60!B2:B61). You should get 60.

Step 3: For total_bill enter: B15 =AVERAGE(Sample_60!B2:B61); B16 =MEDIAN(Sample_60!B2:B61); B17 =STDEV.S(Sample_60!B2:B61). For extra descriptive statistics use =MIN, =QUARTILE.INC(range,1), =QUARTILE.INC(range,3), =MAX, =SKEW and =KURT.

Step 4: Build the 95% t interval for mean total_bill. B18 =B17/SQRT(B14) for the standard error. B19 =T.INV.2T(0.05,B14-1) for the critical t value. B20 =B19*B18 for the margin of error. Put the lower endpoint in D21 with =B15-B20 and the upper endpoint in E21 with =B15+B20.

Step 5: For tip enter B22 =AVERAGE(Sample_60!C2:C61), B23 =STDEV.S(Sample_60!C2:C61), B24 =B23/SQRT(B14), B25 =T.INV.2T(0.01,B14-1), B26 =B25*B24, D27 =B22-B26 and E27 =B22+B26. This produces a 99% t interval.

Step 6: For the smoker proportion enter B28 =COUNTIF(Sample_60!E2:E61,"Yes"), B29 =B28/B14, B30 =NORM.S.INV(0.975), B31 =SQRT(B29*(1-B29)/B14), B32 =B30*B31, D33 =B29-B32 and E33 =B29+B32. Format B29 and D33:E33 as percentages.

Step 7: Sample-size planning for the mean. In the planning table use margin e = 2, confidence = 95%, and estimated σ = the sample SD from B17. Required n =ROUNDUP((NORM.S.INV(0.975)*B17/2)^2,0). Always round sample-size calculations UP.

Step 8: Sample-size planning for the smoker proportion. Using the sample p, enter =ROUNDUP((NORM.S.INV(0.975)^2*B29*(1-B29))/(0.05^2),0). Then repeat with 0.50 in place of B29 for the conservative design.

Step 9: Create a histogram. Select Sample_60!B1:B61, choose Insert > Insert Statistic Chart > Histogram. If needed, right-click the horizontal axis, choose Format Axis, and use a bin width of about \$5. Add the title 'Sample of 60 Restaurant Bills'.

Step 10: Benchmark only after finishing the sample analysis. Full-data mean total bill: =AVERAGE(Full_Data!B2:B245). Full-data mean tip: =AVERAGE(Full_Data!C2:C245). Full-data smoker proportion: =COUNTIF(Full_Data!E2:E245,"Yes")/COUNTA(Full_Data!E2:E245). Compare these benchmarks with your sample confidence intervals.

Final descriptive-statistics results for Sample_60

Statistic	Total bill (\$)	Tip (\$)
n	60	60
Mean	19.5663	3.0550
Median	17.6950	3.0000
Sample SD	8.7910	1.5906
Variance	77.2814	2.5301
Minimum	5.75	1.00
Q1	13.36	2.00
Q3	23.49	3.50
Maximum	50.81	10.00
Skewness	1.1562	1.9594
Kurtosis	1.6847	5.8613

Interpretation: both variables are right-skewed, especially tip. This matters for very small samples; however $n = 60$ is large enough for the sampling distribution of the sample mean to be reasonably well approximated by the Central Limit Theorem in this classroom exercise.

Confidence-interval results

Quantity	Estimate	Critical value	SE	Margin	95% / 99% interval
Mean total bill	\$19.5663	$t^* = 2.0010$	1.1349	2.2710	(\$17.2954, \$21.8373) - 95%
Mean tip	\$3.0550	$t^* = 2.6618$	0.20535	0.54659	(\$2.5084, \$3.6016) - 99%
Smoker proportion	0.4500	$z^* = 1.9600$	0.06423	0.12588	(0.3241, 0.5759) - 95%

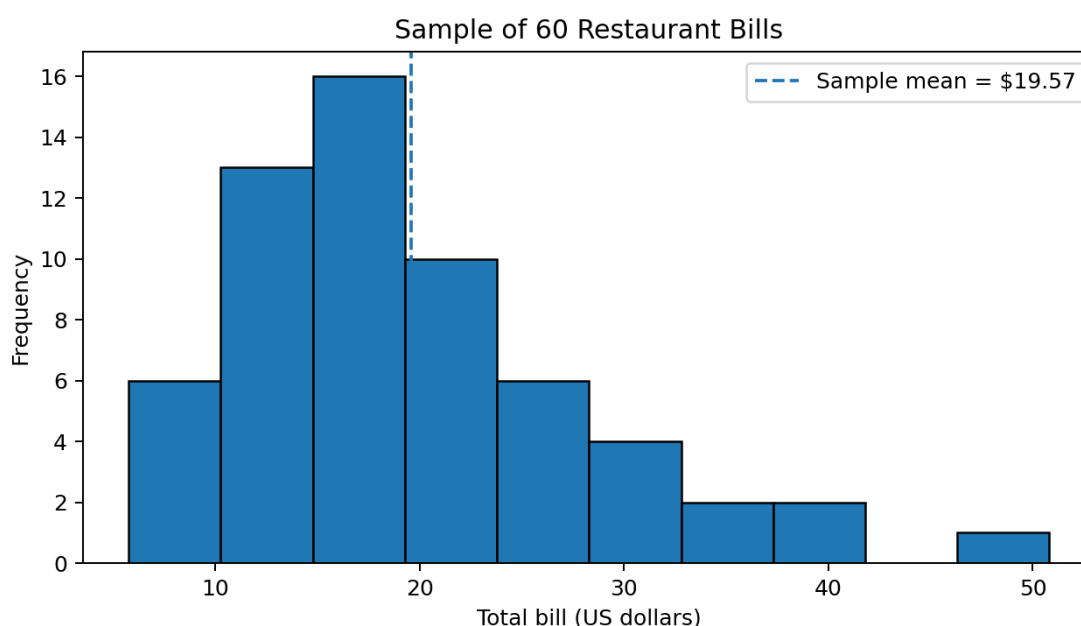
Large-sample check for the smoker interval: $np = 60(0.45) = 27$ and $n(1-p) = 33$, so both exceed 5. The normal proportion interval used in Lecture 7B is therefore appropriate.

Sample-size planning results

Planning objective	Inputs	Required n
Mean total bill within $\pm \$2$ at 95%	$z^* = 1.96$, estimated $\sigma = 8.791$	75
Smoker proportion within $\pm 5\%$ at 95% - sample p	$p = 0.45$	381
Smoker proportion within $\pm 5\%$ at 95% - conservative	$p = 0.50$	385

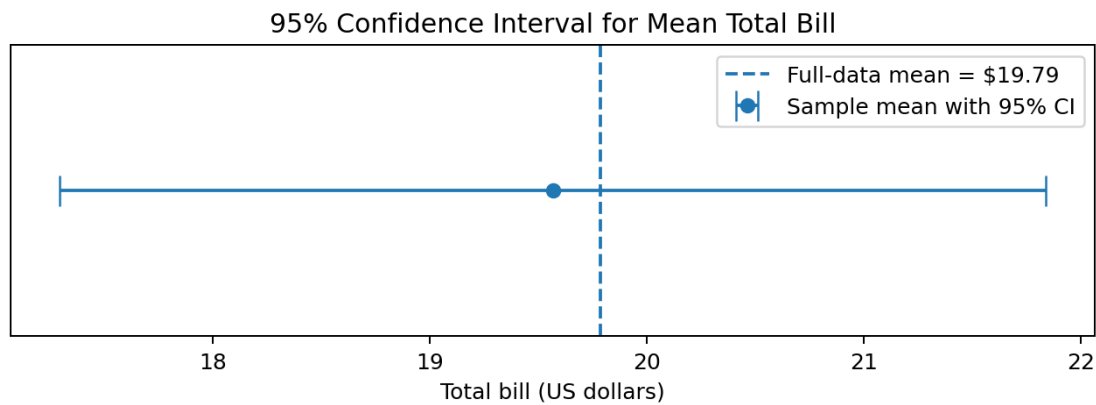
The conservative proportion design is only slightly larger here because the sample estimate $p = 0.45$ is already close to 0.50, where $p(1-p)$ is maximized.

Figure 1. Distribution of total bills in Sample_60



The sample contains many bills around the mid-teens to low twenties, with a longer upper tail and several larger bills. The sample mean is about \$19.57. The histogram is not perfectly normal, but with $n = 60$ the t interval for the mean is supported by the Central Limit Theorem provided the observations can be treated as a random sample from the relevant frame.

Figure 2. Mean total bill confidence interval and full-data benchmark



The full 244-record mean is \$19.7859, which lies comfortably inside the sample 95% confidence interval (\$17.2954, \$21.8373). This is exactly the kind of behaviour confidence intervals are designed to produce repeatedly: the interval is constructed from a sample and aims to cover the population parameter.

Benchmark comparison with Full_Data

Parameter / benchmark	Full-data value	Sample interval	Inside interval?
Mean total bill	\$19.7859	(\$17.2954, \$21.8373)	Yes
Mean tip	\$2.9983	(\$2.5084, \$3.6016)	Yes
Smoker proportion	0.3811	(0.3241, 0.5759)	Yes

Important interpretation: Full_Data is an observed data frame, not the entire universe of restaurant transactions. The valid inferential population is the population represented by the random-sampling frame - transactions comparable to those served by this food server in that restaurant and period. It would be inappropriate to generalize these intervals automatically to all restaurants, all cities, or all customers.

Final Check

For every confidence interval, report: (1) the point estimate, (2) confidence level, (3) standard error, (4) critical value, (5) margin of error, (6) lower and upper confidence limits, (7) sample size, and (8) a context-specific interpretation. For sample-size planning, always round UP.

PART 3 - Pearson MyLab

1. Complete **Quiz #4** on MyLab which has been released on **Tuesday (22-Sep-2026)** and will be **due by the Tuesday on 6-Oct-2026 at 5:00pm AWST**.

Note: If you are having issues with obtaining access to MyLab, please re-read the instructions (available in the welcome topic). If you are still unsuccessful, please send an email to the teaching team explaining the issue. It is important that you complete MyLab tasks on a weekly basis since the final e-test in this unit will be completed using MyLab.